

Exam Date & Time: 08-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Cyber Law and Ethics [CSE-4094]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Government implements a new surveillance program to prevent terrorism. While the program is effective in reducing incidents, it also raises privacy concerns for citizens. Analyze the ethical implications of this program from both a deontological and teleological perspective. Discuss potential solutions that might balance security with individual rights. (5)
 - A)
 - B) You are developing an AI program designed to help with hiring decisions. However, you discover that the program perpetuates unconscious biases based on the data it was trained on.
 - a. Analyze the ethical considerations of AI development and the potential for bias. Discuss how training data can influence AI decision-making and the implications for fairness and equality. (5)
 - b. Evaluate your responsibility as the developer and propose solutions for mitigating bias in the AI program. Consider strategies for diversifying training data, incorporating human oversight, and ensuring ethical guidelines for AI development.
- 2) In today's globalized world, cultural diversity and technological advancements pose both challenges and opportunities for value formation. Analyze the roles that education and family can play in helping individuals develop a strong moral compass in this complex environment. (5)
 - A)
 - B) Online communities often require users to share personal information to create profiles and participate in discussions. Analyze the potential trade-offs between user privacy and fostering a sense of community within online platforms. Suggest strategies that online communities can implement to mitigate privacy risks while still encouraging user engagement. (5)
- 3) The Internet of Things (IoT) is rapidly expanding, connecting billions of devices to the internet. Analyze the security vulnerabilities associated with IoT devices and how cybercriminals can exploit them. Discuss strategies to improve the security of IoT devices and make them less susceptible to cyberattacks. (5)
 - A)
 - B) a. A group of hackers breaches the servers of a powerful corporation known for environmental destruction. They leak internal documents exposing the company's illegal activities and environmental damage. While the hack raises awareness and potentially

forces positive change, it also raises ethical concerns about hacking and the legality of leaked information.

b. Analyze the ethical implications of hacktivism. Consider the potential benefits of exposing wrongdoing against the damage caused by illegal hacking activities.

c. Discuss the legality of the hacktivists' actions and the potential consequences for the corporation and the hacktivists themselves.

d. Evaluate the effectiveness of hacktivism as a tool for social justice and suggest alternative approaches for addressing environmental concerns.

- 4) The patent system is a cornerstone of technological advancement in engineering. Analyze the positive and negative impacts of patents on the field of engineering. Consider how patents can incentivize innovation while also potentially hindering collaboration or access to new technologies. (5)
- A)
- B) Open-source software development has emerged as a significant force in the engineering field. Analyze the potential advantages and disadvantages of open-source software compared to software protected by patents. Consider factors like innovation, collaboration, and commercial viability. (5)
- 5) An engineering firm designs a revolutionary bridge using a unique suspension system. Explain the concept of copyright and analyze how it can be significant for the engineering firm in this scenario. Discuss the limitations of copyright protection for engineering designs and how the firm can potentially extend its intellectual property rights. (5)
- A)
- B) The concept of the public domain plays a crucial role in copyright law. Analyze the significance of public domain works for creativity and access to information. Discuss the factors that determine when a work enters the public domain and how it differs from copyrighted material. Provide examples of how public domain works can be used for creative purposes. (5)

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